

UNIVERSAL TICKETS — PBL EXPO POSTER SUMMARY

Course: 25CS1302E • **Team 2 (Section 9):** V. Hemanth, K. Chaitanya, CH. Vivek • **Guide:** Dr. Prasanthi

1. Problem Statement & Objectives

Live entertainment ticketing requires atomic seat allocation, sub-second search, and role verification. Universal Tickets combines relational ACID transactions with TF-IDF cosine similarity search.

2. Database & Schema (CO1)

MySQL 8.0 with 8 3NF normalized tables: users, organizer_verifiers, organizer_profiles, venues, events, bookings, payments, cancellations. Enforces CHECK constraints, B-Tree indexes, CTEs, and Window Functions.

3. Polyglot Microservices (CO3, CO4, CO5)

- **Spring Boot (8080):** Core transactional engine, JPA/Hibernate.
- **FastAPI (8000):** TF-IDF Semantic Search + OpenAPI Swagger.
- **Node.js (5001):** User activity streaming.
- **React 18 SPA (5173):** Vite client with dark/light mode.

4. Security & Authentication (CO3)

- BCrypt salted password hashing.
- HMAC-SHA512 signed JWT access tokens.
- Multi-tier RBAC for Customer, Organizer, Verifier, Admin.

5. DevOps & CI/CD (CO6)

- Multi-stage Docker containerization & docker-compose.
- GitHub Actions automated Maven/Pytest/Node build pipeline.
- Spring Boot Actuator health & telemetry metrics.

6. Verified Live Entity Counts

Users: 31 • **Venues:** 40 • **Events:** 30
Bookings: 50 • **Payments:** 50 • **Refunds:** 6